Haotong Luan

Shenzhen, China

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Summary

Third-year automation undergraduate working on underactuated robotic hands, environment-adaptive grasping, and mechanism design. Experience spans gripper development, visual-servoing control, and hardware prototyping, with interests in VLMs, VLAs, and WAMs for reliable robotic manipulation.

Education

Shenzhen Technology University – Future Technology School — *Shenzhen, China*

B.Eng. in Automation -- Third-year Undergraduate — *Sep. 2024 – Present*

- GPA: 84.17/100. Research focus: underactuated grippers, environment-adaptive robotic hands, and mechanism design.

Tsinghua University | Shenzhen X-Institute — *Shenzhen, China*

P-SRT Program -- Joint Training in Underactuated Robotics — *Sep. 2025 – Present*

- Advisor: Associate Prof. Wenzeng Zhang. Current stage: P-SRT; program pathway: X-idea → P-SRT → E-SRT → ORIC → SURF.

Research Experience

Shanghai AI Laboratory — *Remote*

Research Assistant -- Visual-Servoing Control and End-Effector Design — *Apr. 2026 – Present*

- Working with Researcher Yuan Gao on visual-servoing control for robotic-arm grasping; responsible for the mechanical design of the arm-mounted gripper.

Shenzhen X-Institute, Tsinghua University — *Shenzhen, China*

P-SRT Researcher -- Underactuated Robotic Grippers — *Sep. 2025 – Present*

- Independently proposed the dual-slide adaptive gripper concept and conducted mechanism analysis and feasibility studies under Associate Prof. Wenzeng Zhang.
- Designed linear-motion linkages, parallelogram mechanisms, and idle-stroke transmissions for reliable single-actuator grasping. Led DS Hand (first author, RAAI 2025); collaborated on idle-stroke and GLINT grippers.

Shenzhen Loop Area Institute — *Shenzhen, China*

Quantum & AI Winter School — *Jan. 2026*

- Studied MLP/LSTM/Transformer-based quantum variational circuit simulation with Prof. Peng Zhang (Tianjin University), including state fidelity optimization, mixed-state simulation, and physics-constrained training.

Publications

* Corresponding author. Submitted manuscripts are listed separately.

Journal Articles

[J1] Validation-Aligned Coreset Selection for Budgeted Few-Shot Classification

Haotong Luan, Xi Yu, Anran Lu, Keyi Chen, Jianwu Chen — *Scientific Reports 2026*

Accepted for publication.

Conference and Workshop Papers

[P1] DPS Hand: A Dual-Parallelogram and Slot-Guided Gripper for Adaptive Grasping and Environmental Scooping

Haotong Luan, Haokai Ding, Tao Yang*, Wenzeng Zhang* — *IEEE CASE 2026*

Accepted; RAS major conference; Shenyang, China, Aug. 17–21, 2026.

[P2] PlaceRecover: Layout-Invariant Point Cloud Recovery for Robust 3D Perception under LiDAR Placement Shifts

Benwu Wang, Zihang Wang*, Wenkai Zhu, **Haotong Luan**, Dong Kong, Binghao Wang, Yiming Zhou, Kailin Lyu, Teng Yan, Haoyu Wang, Yaohua Liu, Qiang Zhang — *IEEE/RSJ IROS 2026*

Accepted; top robotics conference; Pittsburgh, PA, USA, Sep. 27–Oct. 1, 2026.

[P3] LitPlan: A Replayable Workflow Contract for Scientific Literature Agents

Haotong Luan — *Workshop NeurIPS 2026*

Sydney, Australia, Dec. 12, 2026.

[P4] CSD: Conformal Selective Diagnosis for Reliable Bearing Fault Identification Under Domain Shift

Xi Yu, Anran Lu, Keyi Chen, **Haotong Luan***, Yuan Gao* — *IEEE SMC 2026*

Accepted; CCF C; Bellevue, WA, USA, Oct. 4–7, 2026. Corresponding author.

[P5] Domain-Semantic Subspace Stacking for Financial Fraud Detection

Haotong Luan, Minmin Chen, Zhenhui Lin, Haokai Ding, Yiwei Sheng* — *ICIC 2026*

Accepted; CCF C; Toronto, Canada, Jul. 22–26, 2026.

[P6] An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability

Haotong Luan, Wenzeng Zhang*, Tao Yang* *RAAI 2025*

Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423404.

[P7] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Yizhe Chen, **Haotong Luan**, Haokai Ding, Wenzeng Zhang* *RAAI 2025*

Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423397.

[P8] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

Haokai Ding, **Haotong Luan**, Tao Yang*, Wenzeng Zhang* *ICCMA 2025*

Paris, France, Nov. 24–26, 2025. DOI: 10.1109/ICCMA67641.2025.11369602.

Journal Manuscripts Submitted / Under Review

[M1] Manuscript submitted to Advanced Intelligent Systems

Haotong Luan, Jie Zhang, Yuan Gao *Submitted, 2026*

Intelligent systems and robotics; JCR Q1.

Patents Pending

3 invention patents and **1 utility model patent** filed and under review.

Honors & Awards

2026 **AIR Summer School Admission**, Institute for AI Industry Research, Tsinghua University.

2026 **CASE 2026 IEEE RAS Travel Support** (Travel Grant).

2026 **National-Level Project, Principal Investigator**, College Student Innovation and Entrepreneurship Training Program: "Zero-Lash Smart Grasping".

2026 **Provincial-Level Project, Member**, College Student Innovation Training Program: "Multi-Core Fiber Dynamic Optical Trap for Single-Cell Pathology".

2026 **SDG Global Summer School Selection** (tuition waived), Zhejiang University.

2026 **Quantum & AI Winter School Selection**, youngest of only 9 undergraduates among 70+ participants.

2024–2025 **Dean's Special Award for Rising Talent, Zhenxiong**, Future Technology School, SZTU.

2025 **Crystal Prize for Scientific & Technological Innovation**, top 10 across SZTU.

2024, 2025 **Outstanding Student Cadre**, Shenzhen Technology University.

2025 **Excellence Prize**, "Challenge Cup" Student Entrepreneurship Competition, university round.

2025 **Excellence Prize**, "Challenge Cup" Extracurricular Academic Science and Technology Competition, university round.

2025 **Outstanding Volunteer**, 15th National Games of China and 12th National Games for Persons with Disabilities / 9th National Special Olympic Games.

Professional Service & Leadership

2026 **Conference Reviewer**, IEEE-RAS International Conference on Humanoid Robots (Humanoids), a top conference in humanoid robotics.

2026 **Conference Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), a top robotics conference.

2026 **Workshop Reviewer**, IAB @ NeurIPS 2026 Workshop.

2024–Present **Head of Organization Department**, Future Technology School, SZTU (since Dec. 2024).

2024–2025 **Class Monitor & Youth League Secretary**, Sep. 2024–Oct. 2025.

Skills

Mechanism Design Parallel-jaw coupled grippers, linear linkages, parallelogram linkages, underactuated and tendon-driven hands; SolidWorks (modeling, assembly, kinematic simulation); Bambu Lab 3D printing; prototyping and grasping experiments.

Programming Introductory knowledge of Python, PyTorch, and CUDA; writing basic training scripts and running simple model training.

Writing & Figures Familiar with paper writing and scientific figure preparation for leading conferences such as AAAI, NeurIPS, and IROS; using AI to organize literature and assist with manuscript drafting and editing.

AI-assisted Research Using AI to refine experiment plans, support code implementation and debugging, analyze results, and inform subsequent experiments.

Languages Chinese (native); English (CET-4: 580; IELTS scheduled Dec. 2026).